Analyzing Electricity Consumption Patterns Using the London Smart Meters Dataset

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Abstract—The growing demand for electricity in urban areas presents significant challenges for efficient energy management and sustainability, aligning with the United Nations Sustainable Development Goal 7 (SDG7) of ensuring access to affordable and clean energy. This study aims to analyze household electricity consumption patterns and develop predictive models to support smarter energy usage.

Using the London Smart Meters dataset, which includes high-resolution electricity usage data from approximately 5,500 households recorded at 30-minute intervals, we employ big data analytics techniques to handle large-scale data efficiently. Apache Spark is utilized for data processing, while machine learning models are applied for time-series forecasting of electricity consumption.

The analysis explores daily, weekly, and seasonal consumption trends, identifies peak demand periods, and detects inefficiencies in energy usage. The results demonstrate the potential of datadriven approaches to improve energy efficiency and enable proactive energy management strategies.

These findings provide valuable insights for policymakers, energy providers, and urban planners to optimize electricity usage and reduce energy waste. This study highlights the important role of artificial intelligence and big data analytics in advancing sustainable energy solutions and supporting global sustainability goals.

Index Terms—Smart Meters, Electricity Consumption, Big Data Analytics, Apache Spark, Machine Learning, Time-Series Forecasting, Energy Efficiency, Sustainable Development Goal 7 (SDG7)

I. Introduction

The rapid growth of urban populations and technological advancements has significantly increased electricity consumption, posing major challenges for efficient energy management and environmental sustainability. This issue aligns with the United Nations Sustainable Development Goal 7 (SDG7), which aims to ensure access to affordable, reliable, and clean energy for all. Beyond increased demand, inefficient energy usage contributes to higher operational costs, resource depletion, and environmental impacts, highlighting the need for intelligent energy management solutions.

Smart meters have emerged as a powerful technology for monitoring electricity consumption by providing highresolution, real-time usage data at the household level. However, the large volume and complexity of this data make it difficult to analyze using traditional methods, limiting its potential for actionable insights.

This study aims to analyze electricity consumption patterns and develop predictive models to improve energy efficiency and support data-driven decision-making. We hypothesize that electricity usage patterns exhibit identifiable temporal trends and inefficiencies that can be effectively captured and predicted using advanced data analytics and machine learning techniques.

To achieve this, the study utilizes the London Smart Meters Dataset, which contains detailed electricity usage data from approximately 5,500 households recorded at 30-minute intervals. Apache Spark is employed for scalable big data processing, while machine learning models are used for timeseries forecasting and pattern detection.

The findings of this study are expected to provide valuable insights for policymakers, energy providers, and urban planners, enabling more efficient energy distribution, reduced waste, and improved sustainability practices. Ultimately, this research demonstrates how big data analytics and artificial intelligence can be leveraged to address modern energy challenges and contribute to global sustainability goals.

II. Problem Statement

The rapid growth in electricity consumption and the widespread deployment of smart meters have led to the generation of massive volumes of high-resolution energy data. While this data holds significant potential for improving energy management, it also introduces substantial challenges related to scalability, complexity, and real-time analysis. Traditional data processing techniques are often inadequate for handling such large and dynamic datasets, limiting the ability to accurately identify consumption patterns, forecast demand, and detect inefficiencies.

This limitation creates a critical gap in effectively utilizing smart meter data for informed decision-making. Without advanced analytical approaches, energy systems are prone to increased energy waste, higher operational costs, and reduced overall efficiency, ultimately hindering progress toward sustainable energy goals.

To address these challenges, there is a growing need for scalable big data frameworks and intelligent machine learning models capable of extracting actionable insights from complex energy datasets. This study aims to bridge this gap by leveraging big data analytics and predictive modeling techniques to enhance energy efficiency, improve demand forecasting, and support data-driven sustainable energy management strategies.

III. RESEARCH QUESTION

To what extent can big data analytics and machine learning models applied to smart meter data improve the understanding of electricity consumption patterns and enable accurate prediction for enhanced energy efficiency?

IV. DATASET

This study utilizes the London Smart Meters Dataset [1], a publicly available dataset hosted on Zenodo, which contains detailed electricity consumption records from approximately 5,500 households in London. The data is collected at 30minute intervals over multiple years, resulting in a large-scale, high-resolution time-series dataset suitable for advanced data analytics and machine learning applications.

The dataset captures fine-grained variations in electricity usage across different households, enabling in-depth analysis of consumption behavior over time. Its real-world nature enhances the reliability of the study and allows for practical evaluation of data-driven energy management solutions.

Dataset Characteristics:

- Number of records: Approximately 160 million observations
- Time resolution: 30-minute intervals
- Coverage period: Multiple years • Number of households: Approximately 5,500 *Columns (Features):*
- Household ID: Unique identifier assigned to each household
- Timestamp: Date and time of each electricity reading
- Energy Consumption (kWh): Amount of electricity consumed per interval

Data Importance:

This dataset provides high-resolution insights into electricity consumption behavior, enabling comprehensive analysis and supporting data-driven decision-making. It facilitates:

- Analysis of daily and seasonal consumption patterns
- Identification of peak demand periods
- Understanding long-term consumption trends across multiple years
- Supporting the development of predictive models in later stages of the project

V. RELATED WORK

Building on these foundations, recent literature emphasizes the necessity of rigorous preprocessing for smart meter data, which is often characterized by high volatility and noise. Studies by Rehman and Iqbal (2025) and Moustafa et al. (2024) highlight that techniques such as outlier removal, normalization, and handling missing values are crucial for improving model convergence. While traditional statistical models like ARIMA and Prophet offer interpretability, they often struggle with the non-linear nature of energy consumption. In contrast, deep learning architectures like LSTM and Seq2Seq have shown superior performance in capturing long-term temporal dependencies through specialized gating mechanisms.Similarly, a machine learning framework for residential demand estimation incorporated smart meter, climate, building, and socioeconomic data across 58,000 households, highlighting the importance of integrating auxiliary features to enhance prediction performance. Furthermore, Moustafa et al. (2024) conducted a comprehensive comparison of forecasting models including ARIMA, SVM, ANN, CNN, and LSTM, concluding that LSTM achieved superior performance in smart meter data prediction. In another study, Tayseer et al. (2025) proposed a cyber-resilient framework for load forecasting and anomaly detection in smart grids, addressing both prediction accuracy and data reliability. Additionally, Wang et al. (2024) reviewed AI-powered methods for load forecasting, anomaly detection, and demand response, emphasizing their role in improving operational efficiency and energy management. Collectively, these studies demonstrate that combining machine learning techniques with scalable big data frameworks enables accurate modeling of electricity consumption patterns and supports data-driven decision-making. However, there remains a need for more efficient and scalable approaches, which this study aims to address by leveraging advanced predictive models and big data analytics techniques.

Despite the significant advancements in electricity consumption forecasting, existing studies primarily focus on improving prediction accuracy using advanced machine learning and deep learning models. However, many of these approaches face challenges related to scalability and computational efficiency when applied to large-scale smart meter datasets. Additionally, limited attention has been given to integrating big data frameworks with predictive models for real-time analysis. Therefore, there is a clear need for a scalable and efficient approach that combines big data analytics with machine learning techniques to achieve both high accuracy and practical applicability. This study aims to address this gap by leveraging PySpark for large-scale data processing alongside predictive modeling techniques.

Paper	Data	Target	Preprocess	Model	Results
[2]	Low Carbon London dataset	Predicting household electricity consumption	Outlier removal, missing value replacement, hyperparameter tuning with Optuna.	LSTM, Seq2Seq, Prophet	Deep learning models effectively capture temporal patterns in electricity consumption.
[3]	Smart meter electricity data, climate data, building characteristics, socioeconomic data (58,000 households)	Estimating residential electricity demand	filtering of invalid homes; removal of solar/no-AC homes and outliers; exclusion of features with >10% missing values; numerical imputation for missing weather data; date format conversion; OneHotEncoder for categorical variables; StandardScaler for feature standardization.	Ridge; Linear Regression, ElasticNet, Lasso, AdaBoost, Bagging, Gradient Boosting, Random Forest, Extra Trees, MLP KNN	Integrating multiple data sources improves prediction performance.
[4]	Irish Smart Metering Trial dataset (~6,000 households), half-hourly electricity consumption data aggregated to hourly.	Comparing forecasting models for electricity consumption	Half-hourly data were aggregated into hourly intervals, followed by data cleaning, removal of missing/null values, min-max normalization, K-means clustering, and an 80/20 train-test split.	ARIMA, SVM, ANN, CNN, LSTM	LSTM achieved superior performance compared to other models.
[5]	Smart meter data (~2,089 meters, 5 months, Egypt)	Load forecasting and anomaly detection.	K-means clustering, PCA, and cyber-attack simulation.	KMEANS-NN PCA-OCSVM ARIMA, CTREE, MLP, NNETAR	improve prediction accuracy and data reliability.
[6]	review of multiple datasets	Review of AI techniques for load forecasting, anomaly detection, and demand response in power distribution systems, with guidance on model selection.	Not specified	Hybrid models, ML and DL	AI enhances operational efficiency and energy management.

Fig. 1. Summary of Studies

Synthesis and Research Gap:

As summarized in Table 1, a clear trend emerges favoring deep learning architectures—specifically LSTM—for high-dimensional smart meter data due to their ability to outperform models like SVM and CNN in terms of Mean Absolute Percentage Error (MAPE). Furthermore, the integration of multi-dimensional data, including climate and socioeconomic features, significantly reduces prediction bias compared to single-source models. However, a common trade-off observed across these studies is the high computational cost and the challenge of scalability in Big Data environments. This highlights a critical research gap: while models like LSTM provide high accuracy, there remains a need for more efficient and scalable frameworks capable of real-time processing. Consequently, this study leverages advanced big data analytics to bridge the gap between predictive precision and computational efficiency, This study concentrates on scalable machine learning models implemented in PySpark MLLib instead of deep learning architectures like LSTM due to computational constraints and the size of the dataset.

VI. EXPLORATORY DATA ANALYSIS (EDA)

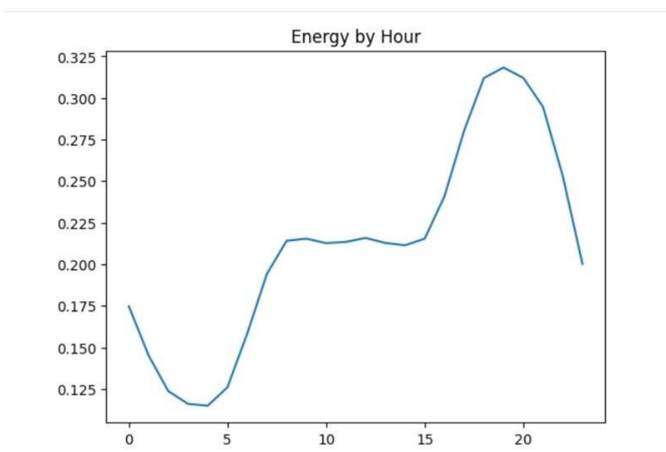

Fig. 2. This chart illustrates the average energy consumption throughout the 24 hours of the day. The energy usage is lowest during the early morning hours, especially between 3 AM and 5 AM, when activity is typically reduced.

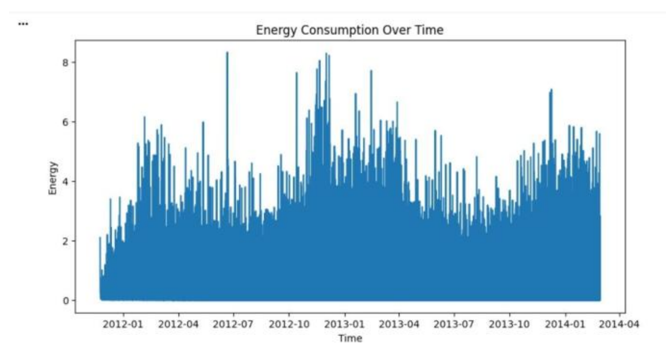

Fig. 3. This time-series chart shows energy consumption over a period from late 2011 to early 2014. The data demonstrates noticeable fluctuations in energy usage over time, with several peaks and variations across different periods. Higher consumption levels appear around late 2012 and early 2013, while lower and more stable levels can be observed during some mid-year periods. Overall, the chart indicates that energy consumption is not constant and may be affected by seasonal changes, usage behavior, and external factors over time.

Consumption begins to increase during the morning and remains relatively stable during the daytime. A significant rise occurs in the evening, with the highest energy consumption observed around 7 PM to 8 PM. This pattern indicates that energy demand increases during evening hours, likely due to higher household or building activity after daytime hours.

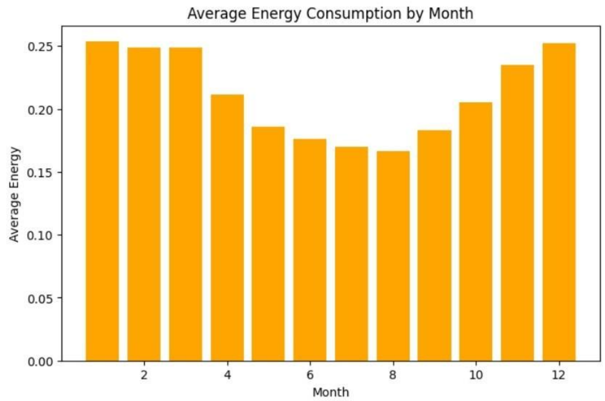

Fig. 4. This chart presents the average energy consumption across the twelve months of the year. The results show a clear seasonal pattern, where energy consumption is relatively high during the beginning and end of the year, especially in January, February, March, and December. In contrast, the lowest average consumption appears during the middle months, particularly July and August. This suggests that energy usage varies by season, possibly due to changes in weather conditions, cooling or heating demand, and user behavior throughout the year.

The exploratory data analysis (EDA) revealed clear temporal patterns in electricity consumption, including daily peaks during evening hours, seasonal variations across months, and long-term fluctuations over the years. These findings directly support the research question by demonstrating that electricity consumption behavior follows identifiable trends that can be modeled and predicted using data-driven approaches. The presence of consistent patterns indicates the feasibility of applying machine learning techniques to improve prediction accuracy and enhance energy efficiency.

VII. Model Building

Data Preprocessing

To facilitate temporal analysis, the timestamp column was transformed into an appropriate timestamp format. To capture time-based patterns of electricity usage, additional temporal characteristics such as hour, day, and month were taken from the timestamp column. Lastly, VectorAssembler in PySpark was used to merge the chosen features into a single feature vector.

To make the energy consumption column compatible with machine learning models, it was transformed into a numerical data type (double). To minimize computational overhead while preserving the overall data distribution, a small representative sample of the dataset was chosen.

Even though outliers were found during the exploratory analysis, they were kept since they probably indicate actual high electricity consumption during periods of peak usage rather than errors or abnormalities in the data.

Linear Regression Model

The Linear Regression model was implemented using PySpark MLLib as a baseline approach for predicting household electricity consumption. It is a simple regression technique used to model the relationship between independent variables and the target variable (energy consumption).

The mathematical formula of Linear Regression is:

$$\hat{y} = \beta_0+\beta_1 x_1 +\beta_2 x_2+\beta_3 x_3$$

where $\hat{y}$ represents the predicted electricity consumption, β_0 is the intercept, and β_1, β_2, and β_3 are the coefficients of the input features. In this project, x_1, x_2, and x_3 represent the temporal features extracted from the timestamp, including hour, day, and month.

The model was trained using temporal features extracted from the timestamp column, including hour, day, and month. These features were selected because electricity usage patterns are strongly influenced by time-based behavior such as daily routines and seasonal variations.

To prepare the data, the selected features were combined into a single feature vector using the VectorAssembler in PySpark. The dataset was then split into 80% training data and 20% testing data to evaluate model performance on unseen data.

Linear Regression was selected as a baseline model to provide a simple benchmark for evaluating the performance of more advanced machine learning models.

Random Forest Regressor

The Random Forest Regressor was implemented as an ensemble learning model to improve prediction accuracy and handle non-linear relationships in the data. It works by constructing multiple decision trees and combining their outputs to produce a more stable and accurate prediction.

The mathematical formula of Random Forest Regressor is:

$$\hat{y} = \frac{1}{N}\sum_{i=1}^{N} T_i(X)$$

where N represents the number of decision trees, $T_i(X)$ represents the prediction of the i-th tree, X represents the input feature vector, and $\hat{y}$ represents the final predicted electricity consumption. The final prediction is obtained by averaging the predictions of all trees.

This model is more robust than Linear Regression because it reduces overfitting and captures complex patterns in electricity consumption data. The same training and testing datasets were used to ensure a fair comparison between all models.Random Forest was selected because of its ability to capture non-linear relationships and reduce overfitting through ensemble learning techniques.

Gradient Boosted Trees (GBT) Regressor

The Gradient Boosted Trees (GBT) Regressor was implemented as an advanced ensemble learning technique that improves prediction performance by sequentially minimizing errors. Each tree is built to correct the mistakes of the previous one, resulting in a stronger predictive model.

The mathematical formula of Gradient Boosted Trees is:

$$\hat{y} = F_0(X) + \sum_{m=1}^{M} \gamma_m h_m(X)$$

where $F_0(X)$ represents the initial prediction, $h_m(X)$ represents the weak learner or tree at iteration m, γ_m represents the learning rate or weight assigned to each tree, M represents the total number of trees, and $\hat{y}$ represents the final predicted electricity consumption.

GBT is effective for capturing complex patterns in largescale datasets and is widely used for regression tasks. The model was trained using the same feature set and dataset split as the other models for consistency in evaluation. GBT was selected due to its strong predictive capability and effectiveness in handling complex patterns within large-scale electricity consumption data.

VIII. Experiments and Results

Experimental Setup

This study was conducted using Apache Spark and PySpark MLLib to efficiently process large-scale smart meter data. The dataset consists of approximately 160 million electricity consumption records collected from around 5,500 households with 30-minute intervals.

Due to the large size of the dataset, a 1% representative sample was used to reduce computational cost while maintaining data distribution. Temporal features including hour, day, and month were extracted from the timestamp column to capture consumption patterns over time.

The dataset was split into 80% training data and 20% testing data. Three regression models were applied: Linear Regression, Random Forest Regressor, and Gradient Boosted Trees (GBT). The models were evaluated using RMSE, MAE, and R^2 metrics.

The Root Mean Squared Error (RMSE) is calculated as:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

where y_i represents the actual electricity consumption value, $\hat{y}_i$ represents the predicted value, and n represents the total number of observations. A lower RMSE value indicates better model performance.

The Mean Absolute Error (MAE) is calculated as:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

where y_i represents the actual electricity consumption value, $\hat{y}_i$ represents the predicted value, and n represents the total number of observations. A lower MAE value means that the model predictions are closer to the actual values. The coefficient of determination, R^2, is calculated as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

where y_i represents the actual electricity consumption value, $\hat{y}_i$ represents the predicted value, and $\bar{y}$ represents the mean of the actual values. A higher R^2 value indicates that the model explains more variance in electricity consumption.

Performance Evaluation

The performance of the models was evaluated using standard regression metrics. RMSE and MAE measure prediction error, while R^2 measures how well the model explains the variance in the data.

The relatively low R^2 values indicate the complexity and variability of household electricity consumption behavior. The models' explanatory ability was limited since they mostly depended on temporal aspects, leaving out significant external elements including weather, occupancy behavior, appliance consumption, and socioeconomic indicators. The low R^2 values are attributed to the use of temporal features only. Future work should incorporate weather, occupancy, and socioeconomic features to improve model explanatory power.

The results indicate that ensemble-based models, particularly Random Forest and GBT, outperformed Linear Regression. This is due to their ability to capture non-linear relationships in electricity consumption data more effectively.

TABLE I

Performance Comparison of Machine Learning Models

Model	RMSE	MAE	R2
Linear Regression	0.291	0.1686	0.031
Random Forest	0.289	0.1669	0.047
GBT	0.287	0.1651	0.056

The results presented in Table I show the performance comparison of the three machine learning models. The Gradient Boosted Trees (GBT) model achieved the best performance with the lowest RMSE and MAE values and the highest R^2 score, indicating higher prediction accuracy for electricity consumption.

In contrast, Linear Regression showed the weakest performance, with higher error values and lower R^2, reflecting its limited ability to capture complex patterns in the data. Overall, ensemble models such as Random Forest and GBT demonstrated better predictive performance compared to Linear Regression.

Feature Importance Analysis

To further interpret model behavior, feature importance analysis was conducted using the GBT model. The results indicated that *hour* and *month* were the most influential predictors, with importance scores of 0.4974 and 0.4950 respectively, while *day* contributed minimally with a score of 0.0077. These findings are consistent with the EDA results, which identified clear daily peaks during evening hours and seasonal variations across months.

TABLE II

FEATURE IMPORTANCE SCORES (GBT MODEL)

Feature	Importance Score
Hour	0.4974
Month	0.4950
Day	0.0077

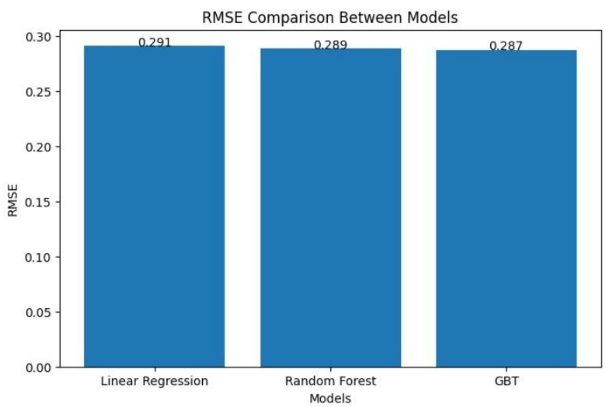

Fig. 5. RMSE Comparison Between Machine Learning Models

Figure 5 shows the RMSE comparison between the machine learning models. It is observed that lower RMSE values indicate better prediction accuracy. The results demonstrate that GBT and Random Forest achieve lower error compared to Linear Regression, indicating improved performance in capturing electricity consumption patterns.

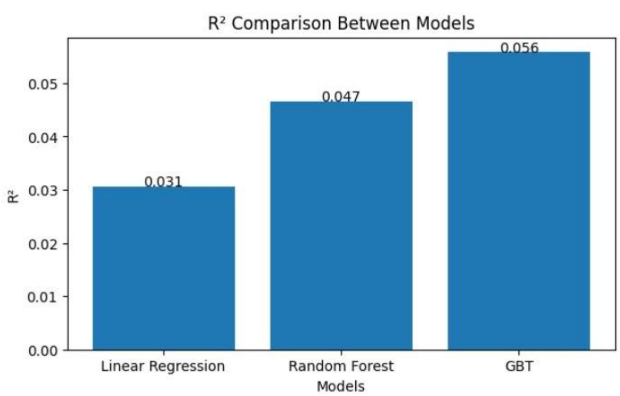

Fig. 6. R^2 Score Comparison Between Machine Learning Models

Figure 6 illustrates the R^2 score comparison between the machine learning models. Higher R^2 values indicate better model fit and stronger ability to explain variance in the data. The results show that GBT achieved the highest R^2 score, followed by Random Forest, while Linear Regression performed the weakest.

Best Model Selection

Based on the evaluation results, the best-performing model was selected according to the lowest RMSE and highest R^2 score. The Gradient Boosted Trees (GBT) model demonstrated better generalization capability and achieved the best trade-off between prediction accuracy and model robustness, resulting in higher accuracy on unseen data.

These results confirm the effectiveness of machine learning and big data analytics in predicting electricity consumption patterns.

IX. CONCLUSION

This study demonstrated the effectiveness of machine learning models combined with big data analytics for predicting household electricity consumption. By using Apache Spark and PySpark MLLib, large-scale smart meter data was efficiently processed and analyzed.

The results showed that ensemble learning models, particularly Gradient Boosted Trees, provided the best predictive performance. Feature importance analysis further revealed that hour and month were the dominant predictors of electricity consumption, validating the temporal patterns identified during EDA. These findings highlight the potential of data-driven approaches in improving energy efficiency and supporting sustainable energy management systems. Although the suggested models produced encouraging results, prediction accuracy and model performance might be further enhanced by adding more external features and using larger training sets. Future work may incorporate weather, demographic, and real-time streaming data alongside advanced deep learning models such as LSTM to improve forecasting accuracy.

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